

Solving the Scale-Up Valley of Death: A Physical Foundation Model for Biomanufacturing

Bioreactor scale-up and process transfer

Scale-up is an inverse problem. Given a process’s oxygen demand, mixing requirements and hydrodynamic limits, determine how a specific reactor must operate. UNIPHY is developing a multi-scale physical foundation model to solve this class of problem through GPU-accelerated simulation, constrained optimization and reusable process specifications.

The physical problem at manufacturing scale

A successful 10 L process does not specify the operating conditions for a 2,000 L vessel. Geometry, impeller configuration, gas delivery and working volume reshape circulation and mass transfer. The engineering task is to preserve the conditions the cells require as those physical conditions change. Validated CFD already provides a foundation for this work; UNIPHY’s target is faster equipment-specific prediction and inverse design. [1]

Predicting conditions inside the production reactor

UNIPHY takes reactor CAD geometry, liquid properties and operating conditions as inputs. Its fluid-dynamics model combines Navier-Stokes flow calculations with transport descriptions to estimate mixing, oxygen transfer and hydrodynamic stress. Consider adapting a 10 L process to a 2,000 L reactor: the inverse-design task evaluates combinations of impeller speed, aeration and working volume against defined process requirements.

The volumetric mass-transfer coefficient, kLa , describes gas-to-liquid oxygen transfer. Local oxygen availability also depends on circulation and cellular consumption. Stress fields identify hydrodynamic exposure; translating that exposure into cell damage requires a calibrated biological relationship. The engineering output is an operating region with explicit assumptions and margins.

Physics across the manufacturing sequence

Capability	Physical model	Design question
Molecular simulation	Coarse-grained molecular dynamics	Which molecular behaviors warrant targeted binding studies?
Bioreactor scale-up	Fluid dynamics and oxygen transport	Which reactor settings satisfy process requirements?
Purification	Porous flow and protein adsorption	Which separation conditions balance recovery and throughput?
Formulation and delivery	Liquid flow and viscosity models	How will the formulation flow during filling and injection?

Molecular kinetics, cell-line performance, purification recovery and formulation stability require distinct validation. Martini simulation time cannot be universally converted into physical binding residence time. [2]

Moving a process between facilities

UNIPHY represents equipment geometry separately from material properties, process requirements and controls. The process specification becomes reusable. A receiving facility supplies a new equipment model; the computational workflow recalculates the flow and transport response and searches for compatible operating conditions.

For Factory A and Factory B, UNIPHY carries forward the process requirements while each equipment configuration receives its own operating solution. Changing the CAD file initiates re-optimization against the receiving reactor's geometry and limits. This is computational process portability: retain the specification, replace the equipment model, recalculate the operating conditions.

The business objective is to reduce repeated engineering work when equipment changes. Each facility still needs a calculation matched to its actual equipment, material properties and biological demand.

Inverse design with explicit uncertainty margins

A reference formulation expresses the engineering objective as J , controls as u , equipment geometry as G , and uncertain material and biological parameters as θ . Equipment constraints define $U(G)$. The optimizer seeks:

$$\begin{aligned} u^* &= \arg \min_{u \in U(G)} J(u; G) \\ \hat{C}(x, t; u, \theta, G) - \delta_C &\geq C_{\min} \\ \hat{S}(x, t; u, \theta, G) + \delta_S &\leq S_{\lim} \end{aligned}$$

The constraints apply across specified modeled positions, times and uncertain parameters θ in Θ . $\hat{C}$ predicts dissolved oxygen; $\hat{S}$ predicts a defined hydrodynamic-stress metric. Experimentally justified margins δ_C and δ_S account for model error. J can encode an engineering objective such as energy demand. This reference formulation defines admissible settings under stated assumptions; verifying the complete operating region requires evaluating the uncertainty set.

Karush-Kuhn-Tucker conditions assess optimality within the mathematical problem. For differentiable convex formulations, satisfying these conditions certifies a global optimum. Bounds on physical prediction error require separate calibration and verification. The result is an operating solution with a defined domain of validity. [3]

Computational speed and manufacturing evidence

UNIPHY reports internal simulations covering 10,000 operating scenarios in hours. This computational throughput targets a concrete bottleneck: evaluating equipment-specific settings before committing to plant trials. Reproducible benchmarks specify hardware, model resolution and solver fidelity; comparisons with physical measurements establish predictive accuracy and actual reductions in development time.

UNIPHY targets rapid computational adaptation between facilities. Completed technology transfer additionally requires the receiving site to establish process performance, transfer analytical methods and complete the relevant qualification, documentation and validation work. Computational recalculation and manufacturing transfer are separate milestones. [4]

Selected references

- [1] Validation of a CFD model for cell culture bioreactors at large scale and its application in scale-up. 2024. PMID 38582408.
- [2] Martini Force Field Initiative. Frequently Asked Questions. Model limitations and interpretation of simulation time.
- [3] Boyd S, Vandenberghe L. Convex Optimization. 2004. Section 5.5.3, KKT optimality conditions.
- [4] WHO. Guidelines on technology transfer in pharmaceutical manufacturing. TRS 1044, Annex 4. 2022.